CSO 101: Computer Programming

Syllabus:

SI No.	Topic	Detailed Content	Contract Hours
1	Introduction to Computing	Definition of Computer, Block Diagram of Computer System, Stored-Programmed Computer, High-Level and Machine-Level Language, Role of Compiler/Interpreter.	1
2	Introduction to the C Programming Language	Definition of Algorithm, Pseudocode, and Flowchart. Brief History of C Language, Structure of C Programming Language, Executing a simple C code.	1
3	Fundamentals of C	Constants, Variables, identifiers, keywords, and Data types, Modifiers, Sizeof Operator, Storage classes.	2
4	Input/Output Functions in C	Printf, Scanf, Unformatted and formatted Input/Output, Type Casting.	1
5	Number Representation in a Computer System	Decimal, Binary, Hex, and Octal Representation and Conversion. Representation of Negative Numbers: Signed Magnitude, 1's Complement, 2's Complement, Floating Point Number Representation, Concept of Biased Exponent.	3
6	Operators and expressions in C	Arithmetic, Relational, Logical, assignment, conditional, increment or decrement, bitwise, special operators, associativity, and precedence of operators.	2
7	Decision making	Different forms of if statements, switch case, continue, and break.	2
8	Looping	for, while, do-while, nested loop, algorithmic problem illustration on loops.	3
9	Arrays	Single and Multi-Dimensional Arrays, Strings, and algorithmic problem illustration on loops and Arrays.	4
	Mid-Term Examination		

10	Functions	Function Definition, Recursive Function, Function Calls, Storage Classes, Scope of Identifiers, Duration of Identifiers. Function Call by Address, Various String Functions.	4
11	Pointers	Introduction to Pointers, Strings, Constant with Pointers: const can be used in four ways with pointers: non-constant pointers to non-constant data, constant pointers to non-constant data (Array), non-constant pointers to constant data, etc.	5
12	Dynamic Memory Allocation	Dynamic Memory Allocation	1
13	Structures & Union	Structures & Union	2
14	Linear Data Structure - Linked List	Linked List, and associated operations - Insertion and Deletion	3
15	Basic Searching and Sorting Algorithms	Linear Search, Binary Search, Insertion Sort, Bubble Sort	2

Grading:

Mid-sem: 30%, End-sem: 40%, Assignments/Quizzes (to be conducted by individual faculty): 10%, Lab (Assignments + Viva): 20%